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RESEARCH

Effectiveness of low-level laser therapy in the rehabilitation of knee meniscus injuries: a randomized, double-blind, placebo-controlled trial

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Abstract

To evaluate the effectiveness of low-level laser therapy (LLLT) as an adjunct to rehabilitation in reducing pain and improving knee function in adults with MRI-confirmed meniscus injuries. In this randomized, double-blind, placebo-controlled trial, 40 adults with unilateral meniscus injuries were assigned to either an LLLT group or a placebo group. All participants received a standardized rehabilitation program over six weeks. The primary outcome was pain intensity measured using a visual analogue scale (VAS). Secondary outcomes included knee function (Lysholm score), range of motion (ROM), and physical performance tests. Both groups showed significant improvements; however, the LLLT group demonstrated greater reductions in pain and improvements in function, mobility, and ROM compared with the control group ($p<0.01$ for all outcomes). The magnitude of improvement exceeded clinically meaningful thresholds, particularly for pain reduction and functional gains. Adjunctive LLLT was associated with greater improvements in pain and functional outcomes compared with rehabilitation alone. LLLT appears to be a safe and potentially effective complementary modality in the conservative management of meniscal injuries.

Keywords Low-level laser therapy · Photobiomodulation · Meniscus injury · Rehabilitation · Randomized controlled trial

Introduction

Knee meniscus injuries are a common cause of knee pain, functional limitation, and early degenerative changes. Degenerative meniscal alterations are highly prevalent, particularly among adults over 40 years of age, and are strongly associated with the progression of knee osteoarthritis (KOA). However, meniscal injuries represent a distinct clinical condition, and findings from osteoarthritis studies cannot be directly generalized without further investigation. Epidemiological studies also indicate that meniscal tears—whether traumatic or degenerative—occur frequently across adulthood, with medial tears reported more often than lateral tears [1].

Conservative rehabilitation remains the preferred first-line management for most meniscal lesions. Structured physiotherapy programs focusing on neuromuscular control, joint mobility, and strengthening have demonstrated meaningful improvements in pain and function among individuals with isolated meniscal pathology or early degenerative lesions [2]. However, symptom improvement is often gradual, prompting many patients to seek adjunctive non-invasive therapies to enhance recovery.

Low-level laser therapy (LLLT), also referred to as photobiomodulation therapy, has gained increasing attention in recent years as a complementary modality in musculoskeletal rehabilitation. LLLT applies low-intensity red or near-infrared light that stimulates mitochondrial activity, enhances cellular metabolism, and modulates inflammatory pathways. These effects are particularly relevant to meniscal tissue, which has limited vascularity and a reduced intrinsic healing capacity. In addition, LLLT may contribute to pain reduction through modulation of nociceptive pathways and improvement of local microcirculation [3]. Moreover, clinical trials in knee osteoarthritis have reported reductions in pain and improvements in joint function and physical performance following LLLT, especially when combined with exercise-based rehabilitation [4].

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Despite preliminary evidence suggesting potential benefits of LLLT for meniscal pathology, the available research remains limited and heterogeneous. Existing studies vary in laser parameters, treatment protocols, and outcome measures, and few high-quality randomized controlled trials have specifically evaluated LLLT in patients with confirmed meniscal injuries [5, 6]. Therefore, this study aimed to evaluate the effectiveness of low-level laser therapy in reducing pain and improving knee function, mobility, and range of motion among adults with MRI-confirmed meniscus injuries undergoing standardized rehabilitation. We hypothesized that adjunctive LLLT would result in greater improvements in pain, function, mobility, and range of motion compared with rehabilitation alone.

Methodology

Study design

This study was a prospective, randomized, double-blind, placebo-controlled clinical trial designed to assess the effects of LLLT on meniscal injury rehabilitation, following prior methodological frameworks used in LLLT trials [7].

Clinical trial number Not applicable.

Participants

A total of 40 individuals (aged 25–50 years) with unilateral knee pain were recruited from outpatient clinics. Inclusion criteria were: (1) diagnosis of a meniscus injury confirmed by magnetic resonance imaging as either a grade-3 attenuation or an intrasubstance tear; (2) knee pain lasting at least six weeks; and (3) ability to participate in rehabilitation exercises. Exclusion criteria included bilateral knee pain, previous history of major knee trauma or joint surgical operation, inflammatory arthropathies (e.g., rheumatoid arthritis), coagulation disorders, systemic diseases (e.g., hemophilia or amyloidosis) and concurrent meniscus-ligament injuries, based on criteria used in the meniscal pathology trial. Participants were stratified by injury type and sex and then randomly assigned (1:1) to receive either LLLT ($n=20$) or placebo therapy ($n=20$) using a computer-generated randomization sequence in which group assignments were hidden in ambiguous envelopes that were sealed and numbered in sequence to prevent selection bias. Both participants and treating therapists remained blinded to group allocation throughout the study. Demographic variables including age, sex and the calculated body mass index (BMI) were recorded at baseline [7, 8].

Interventions

Participants in the intervention group received infrared gallium arsenide (GaAs) laser therapy (904 nm; Irradia Medical Laser, M/D Laser Professional, Stockholm, Sweden) delivered through four infrared diodes with an average output power of 240 mW (4 × 60 mW). The device operated in pulsed mode using frequencies of 2400 Hz (anti-inflammatory frequency) and 700 Hz (healing frequency), consistent with the protocol described by Malliaropoulos et al. [9]. The device operated in pulsed mode rather than continuous-wave mode, and the frequencies of 2400 Hz and 700 Hz represented the pulse repetition rates applied during treatment.

The medial meniscus region was divided into two rows of ten treatment points. The laser was applied in a stationary point-by-point mode, with a spot area of approximately 1 cm^2 and a power density of 0.24 W/cm^2. Each treatment point received 10.5 s of irradiation, corresponding to a dose of 2.52 J per point. The total treatment time was 420 s per session (210 s at 2400 Hz and 210 s at 700 Hz), resulting in a total energy delivery of 100.8 J per knee per session.

Placebo group

Participants allocated to the placebo group received sham irradiation using an identical laser device with inactive diodes. The sham device was visually indistinguishable from the active device, emitted the same indicator lights, and followed the same treatment procedures and duration. Because low-level laser therapy does not normally produce perceptible heat sensations, participants were unable to distinguish active from sham treatment and remained unaware of group allocation throughout the study. The active and sham devices were pre-programmed and coded by an independent researcher; therefore, treating therapists remained unaware of treatment allocation throughout the study. Outcome assessments were performed by an evaluator blinded to group allocation.

Rehabilitation program (Both Groups)

All participants completed a standardized physiotherapy program consisting of mobility, strengthening, and neuromuscular control exercises over a six-week period under the supervision of a licensed physiotherapist.

Outcome measures

Assessments were conducted at baseline and four weeks after completion of the intervention (i.e., week 6).

Primary outcome The primary outcome was pain intensity assessed using a 10-cm Visual Analogue Scale (VAS) [10, 11].

Secondary outcomes

Lysholm Knee Score knee-related function was assessed using the Lysholm Knee Scoring Scale, a validated patient-reported outcome measure evaluating pain, instability, swelling, stair climbing, limping, locking, squatting, and need for support. Scores range from 0 to 100, with higher scores indicating better knee function. The questionnaire was administered by the evaluator, and instructions were provided in Arabic to ensure participant understanding [12].

Range of Motion (ROM) Knee flexion ROM was measured in degrees using a standard universal goniometer. Participants were assessed in a standardized supine position at baseline and post-treatment by the same evaluator to ensure consistency. Three measurements were obtained for each assessment and the average value was used for analysis. Universal goniometry is a widely accepted clinical method with demonstrated reliability and validity for knee ROM assessment in musculoskeletal populations.

Physical performance tests Functional mobility was assessed using a standardized 20-m walking test and a stair-climb test (ascending and descending a flight of 12 steps). Participants were instructed to complete each test as quickly and safely as possible without running. One familiarization trial was performed before testing, followed by two recorded trials, with the best performance used for analysis. Completion time was recorded in seconds. These tests are commonly used measures of functional performance in individuals with knee disorders and have demonstrated acceptable validity, reliability, and responsiveness in musculoskeletal rehabilitation settings [13].

Treatment response classification

Participants were classified as treatment responders if they achieved both a ≥ 30% reduction in pain intensity on the Visual Analogue Scale (VAS) and a ≥ 5-point improvement in the Lysholm Knee Score, a validated patient-reported outcome measure (PROM). These criteria were selected to represent clinically meaningful improvements in both pain and self-reported knee function [10, 11].

Statistical analysis

All analyses followed the intention-to-treat principle. Continuous variables were reported as means and standard deviations (SD). Within-group changes from baseline to post-treatment were evaluated using paired t-tests or Wilcoxon signed-rank tests for non-normally distributed data. Differences in improvement scores between groups were examined using independent-samples t-tests or Mann–Whitney U tests, as appropriate. Categorical variables, including treatment responder status, were analyzed using chi-square tests. Treatment effect size was estimated using Cohen's d, and statistical significance was defined as $p < 0.05$ (two-tailed) [14].

Results

Overall results

Both groups showed significant improvements from baseline across all outcomes. However, the LLLT group demonstrated significantly greater improvements compared to the control group. Pain intensity (VAS) decreased by 3.36 ± 1.39 in the LLLT group compared with 2.08 ± 1.36 in the control group ($p = 0.0053$; Cohen's d = 0.94). The Lysholm score improved by 4.70 ± 2.81 in the LLLT group versus 1.65 ± 1.73 in controls ($p < 0.001$; d = 1.31). Walking time and stair-climb time decreased significantly more in the LLLT group ($p < 0.001$), with a large effect size (d = 1.89–2.08). Knee flexion ROM improved by 6.31 ± 2.84° in the LLLT group compared with 2.70 ± 1.71° in controls ($p < 0.001$; d = 1.54). Table 1 shows the baseline demographic and clinical characteristics of the participants.

Table 1 Baseline characteristics of participants in LLLT and control groups

Variable	LLLT (mean ± SD)	Control (mean ± SD)	*p*-value
Age (years)	36.85 ± 8.37	34.45 ± 7.24	0.338
BMI (kg/m²)	24.77 ± 2.14	25.78 ± 1.58	0.100
Baseline VAS	8.41 ± 0.86	8.45 ± 0.98	0.905
Baseline Lysholm	75.65 ± 4.28	77.50 ± 3.78	0.156
Baseline walking test (s)	84.70 ± 8.94	83.25 ± 11.35	0.656
Baseline stair-climb test (s)	73.35 ± 8.01	78.25 ± 9.10	0.079
Baseline ROM (°)	120.30 ± 5.76	123.60 ± 8.00	0.144

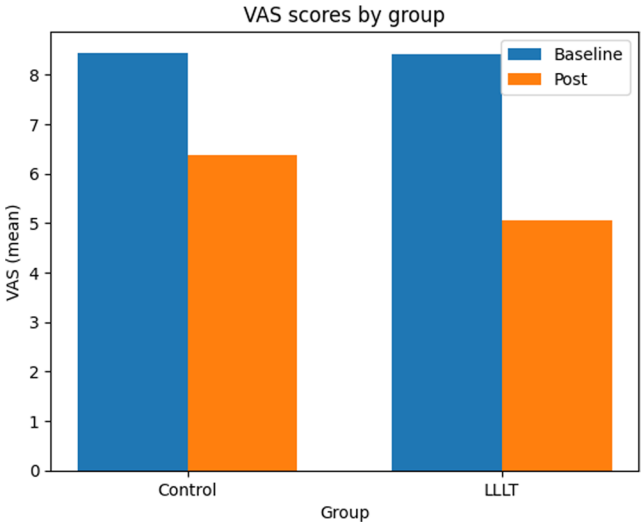


Fig. 1 VAS (Pain) Scores by Group: Baseline vs. Post-Treatment (LLLT vs. Control)

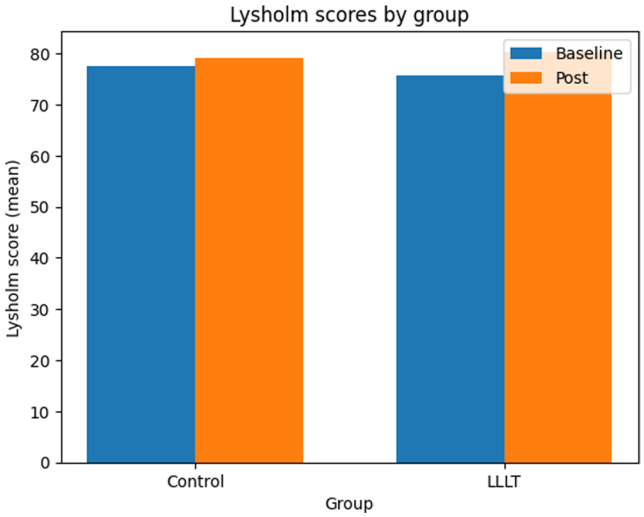


Fig. 2 Lysholm knee function scores by group: Baseline vs. Post-Treatment (LLLT vs. control)

Primary outcomes

Figure 1Comparison of Pain Reduction (VAS) Between LLLT and Control Groups.

Pain scores decreased significantly in both groups. Table 2 summarizes the comparison of pain reduction (VAS) between the two groups. In addition, Fig. 1 illustrates the change in VAS pain scores from baseline to post-treatment in both groups. The LLLT group showed a larger reduction (3.36 points vs. 2.08 points), yielding a between-group p-value of 0.0053, and a large effect size was observed (Cohen's d = 0.94).

Knee function (Lysholm)

Function improved in both groups as shown in Table 3; Fig. 2, but the LLLT arm gained nearly three points more than control (between-group $p < 0.001$; Cohen's d = 1.31).

Secondary outcomes

The LLLT group walked 24.8 s faster post-treatment compared with an 11.2-s improvement in controls ($p < 0.001$; Cohen's d = 1.89). Table 4 shows the walking test performance at baseline and post-treatment. In addition, Fig. 3 presents the walking test performance before and after treatment in both groups.

Stair-climbing test

LLLT reduced stair-climb time by 18.4 s, more than double the 7.8-s reduction in controls ($p < 0.001$; Cohen's d = 2.08), indicating that a large effect size was observed (Table 5; Fig. 4).

Range of Motion (ROM)

ROM increased significantly in both groups as shown in Table 6; Fig. 5, with an additional 3.6° gain in the LLLT arm ($p < 0.001$; Cohen's d = 1.54).

Table 2 Comparison of pain reduction (VAS) Between LLLT and control groups

Group	Baseline VAS	Post VAS	Reduction	*p*-value (within group)
Control	8.45 ± 0.98	6.37 ± 1.16	2.08 ± 1.36	$p < 0.001$
LLLT	8.41 ± 0.86	5.04 ± 1.23	3.36 ± 1.39	$p < 0.001$

Table 3 Comparison of knee function (Lysholm Score) Between LLLT and control groups

Group	Baseline Lysholm	Post Lysholm	Improvement	*p*-value (within group)
Control	77.50 ± 3.78	79.15 ± 3.38	1.65 ± 1.73	$p < 0.001$
LLLT	75.65 ± 4.28	80.35 ± 4.82	4.70 ± 2.81	$p < 0.001$

Table 4 Comparison of walking test performance between LLLT and control groups

Group	Baseline (s)	Post (s)	Change (s)	*p*-value (within group)
Control	83.25 ± 11.35	72.00 ± 14.03	−11.24 ± 5.74	$p < 0.001$
LLLT	84.70 ± 8.94	59.85 ± 12.76	−24.84 ± 8.42	$p < 0.001$

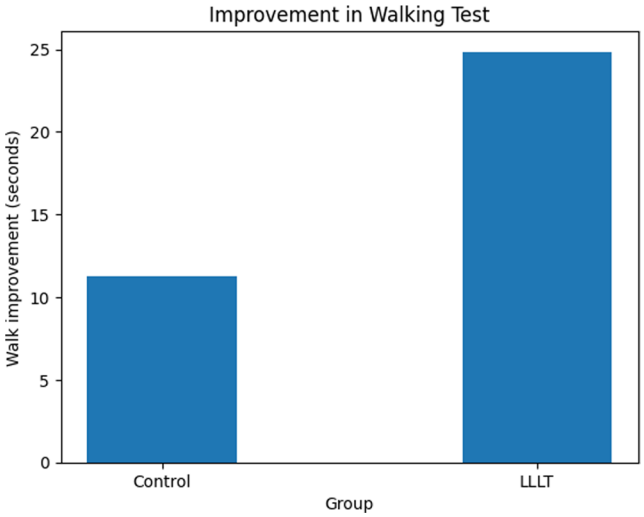


Fig. 3 Walking test performance by group: Baseline vs. Post-Treatment (LLLT vs. Control)

Between-group improvements

Table 7 summarizes the between-group differences in outcome improvements across all measured variables.

The between-group differences remained statistically significant for all outcomes, with 95% confidence intervals favoring the LLLT group and effect sizes suggesting clinically relevant benefits.

Treatment response classification

Table 8 presents the classification of participants according to treatment response in both groups.

Discussion

This randomized, double-blind, placebo-controlled trial investigated the effectiveness of low-level laser therapy (LLLT) as an adjunct to rehabilitation in individuals with knee meniscus injuries. The findings of this study indicate that the addition of LLLT to a standardized rehabilitation program was associated with greater improvements in pain, knee function, physical performance, and range of motion compared with rehabilitation alone. Overall, these findings suggest that LLLT may provide additional therapeutic benefits when incorporated into conservative management strategies for meniscal injuries.

The findings of the present study are consistent with previous research reporting beneficial effects of low-level laser therapy (LLLT) in musculoskeletal conditions. Several studies have demonstrated that photobiomodulation therapy can significantly reduce pain and improve functional outcomes, particularly when combined with exercise-based rehabilitation programs [2, 15]. In addition, randomized controlled trials in individuals with knee osteoarthritis have reported improvements in pain, physical performance, and joint function following LLLT interventions [16, 17]. Although most of the available evidence has focused on osteoarthritis populations, the current study extends these findings to individuals with MRI-confirmed meniscal injuries. This is particularly relevant given the limited number of controlled trials specifically targeting meniscal pathology, thereby contributing to the growing body of evidence supporting the clinical applicability of LLLT in knee rehabilitation.

Beyond pain reduction, the observed improvements in knee function, walking performance, stair-climbing ability, and range of motion are consistent with recent rehabilitation research emphasizing the importance of functional recovery and movement quality in knee-related musculoskeletal conditions. Previous studies have demonstrated that improvements in functional performance, mobility, and biomechanical symmetry are associated with better rehabilitation outcomes and return to activity following knee pathology and surgery [18, 19]. These findings support the clinical relevance of the functional gains observed in the present study.

Recent rehabilitation research has further emphasized the importance of movement quality, neuromuscular control, and exercise-based interventions in optimizing rehabilitation outcomes. Studies have demonstrated that specific exercise

Table 5 Comparison of stair-climb test performance between LLLT and control groups

Group	Baseline (s)	Post (s)	Change (s)	*p*-value (within group)
Control	78.25 ± 9.10	70.48 ± 10.23	−7.77 ± 3.90	$p < 0.001$
LLLT	73.35 ± 8.01	54.94 ± 9.07	−18.41 ± 6.09	$p < 0.001$



Fig. 4 Comparison of stair-climb test performance between LLLT and control groups

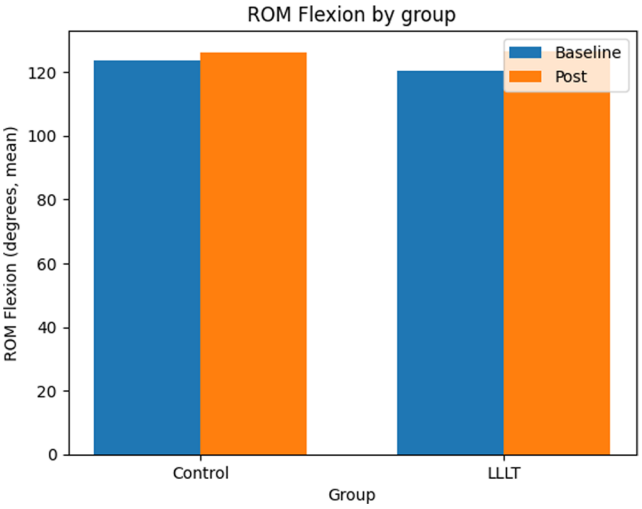


Fig. 5 Range of Motion (ROM) Flexion by Group: Baseline vs. Post-Treatment (LLLT vs. Control)

modifications can significantly influence muscle activation patterns and functional performance, highlighting the value of targeted rehabilitation strategies in musculoskeletal care. These findings support the rationale for combining adjunctive modalities with structured rehabilitation programs to enhance functional recovery which may partly explain the superior functional improvements observed in the LLLT group [20, 21].

Several biological mechanisms may explain the observed benefits of LLLT in the present study. Photobiomodulation has been shown to enhance mitochondrial activity and increase adenosine triphosphate (ATP) production, which can promote cellular repair and tissue regeneration [3]. In addition, LLLT has been reported to modulate inflammatory processes by reducing pro-inflammatory mediators and enhancing local microcirculation, which may contribute to pain reduction and improved functional recovery [6]. These effects are particularly relevant in meniscal tissue, which is characterized by limited vascularity and a reduced intrinsic healing capacity. Furthermore, LLLT may influence nociceptive pathways, leading to decreased pain perception and facilitating participation in rehabilitation exercises. Collectively, these mechanisms provide a plausible explanation for the superior improvements observed in the LLLT group [22, 23].

While the findings suggest additional benefits of LLLT, alternative explanations should be considered. Both groups participated in a structured rehabilitation program, which is known to improve pain and functional outcomes in individuals with meniscal injuries. Therefore, part of the observed improvements may be attributed to the effects of exercise-based rehabilitation rather than the laser intervention alone. In addition, placebo effects and patient expectations may have influenced subjective outcomes such as pain perception. However, the randomized, double-blind, placebo-controlled design of the study reduces the likelihood that these factors fully account for the observed between-group differences.

The findings of this study have potential clinical implications for the conservative management of meniscal injuries. Given the non-invasive nature of LLLT and its favorable safety profile, it may be considered as a complementary modality alongside standard rehabilitation programs. The observed improvements in pain, function, and physical performance suggest that incorporating LLLT could enhance rehabilitation outcomes and potentially accelerate recovery. This may be particularly beneficial for individuals seeking non-surgical treatment options or those with limited tolerance for more invasive interventions.

Limitations

Several limitations should be acknowledged when interpreting the findings of this study. First, the relatively small sample size may limit the generalizability of the results and reduce statistical power. Second, the short follow-up period prevents conclusions regarding the long-term sustainability of the observed improvements. Third, the study was conducted at a single center, which may limit external validity

Table 6 Comparison of Range of Motion (ROM) improvement between LLLT and control groups

Group	Baseline (°)	Post (°)	Increase (°)	*p*-value (within group)
Control	123.60 ± 8.00	126.30 ± 8.12	2.70 ± 1.71	$p < 0.001$
LLLT	120.30 ± 5.76	126.61 ± 5.31	6.31 ± 2.84	$p < 0.001$

Table 7 Between-group differences in outcome improvements (VAS, Lysholm, Walking, Stair-Climb, ROM)

Outcome	LLLT Δ (mean ± SD)	Control Δ (mean ± SD)	Difference	95% CI	*p*-value	Effect size
Pain (VAS) reduction	3.36 ± 1.39	2.08 ± 1.36	1.29	0.41 to 2.17	0.0053	0.94
Lysholm score gain	4.70 ± 2.81	1.65 ± 1.73	3.05	1.55 to 4.55	$p<0.001$	1.31
Walking time reduction (s)	−24.84 ± 8.42	−11.24 ± 5.74	−13.60	8.97 to 18.23	$p<0.001$	1.89
Stair-climb time reduction (s)	−18.41 ± 6.09	−7.77 ± 3.90	−10.65	7.35 to 13.94	$p<0.001$	2.08
ROM flexion increase (°)	6.31 ± 2.84	2.70 ± 1.71	3.61	2.10 to 5.13	$p<0.001$	1.54

Table 8 Treatment response classification in the LLLT and control groups

Group	Treatment Responders (*n*, %)	Non-responders (*n*, %)
Control	14 (70%)	6 (30%)
LLLT	17 (85%)	3 (15%)

and applicability to broader populations. In addition, only one set of laser parameters was evaluated; therefore, the findings cannot be generalized to other treatment protocols or dosages. Importantly, the trial was not prospectively registered in a public clinical trial registry. The lack of prospective registration may reduce transparency and limit external verification of prespecified outcomes and analyses. Finally, although both groups followed an identical rehabilitation program, the combined effects of exercise and LLLT make it difficult to fully isolate the independent contribution of the laser intervention.

Despite these limitations, the findings of this study provide important insights into the potential role of LLLT in meniscal rehabilitation.

In summary, the findings of this study suggest that low-level laser therapy (LLLT) may provide additional benefits when used as an adjunct to rehabilitation in individuals with meniscal injuries. While the results are promising, they should be interpreted with caution in light of these limitations. Further high-quality studies with larger sample sizes and longer follow-up periods are warranted to confirm these findings and to establish optimal treatment parameters.

Conclusion

Low-level laser therapy (LLLT) may be a useful adjunct to rehabilitation in individuals with knee meniscus injuries. Compared with placebo treatment, adjunctive LLLT was associated with greater improvements in pain, knee function, physical performance, and range of motion. Between-group differences favored the LLLT group across the assessed outcomes, with moderate-to-large effect sizes and 95% confidence intervals indicating potentially meaningful clinical benefits. However, these findings should be interpreted cautiously due to the relatively small sample size, lack of prospective trial registration, and short follow-up duration. Further well-designed randomized controlled trials with larger samples and longer follow-up periods are needed to confirm these findings and establish optimal treatment parameters.

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Author contributions H.A. was solely responsible for the conception and study design, participant recruitment, data collection, and experimental implementation. H.A. performed the statistical analysis, interpreted the findings, and prepared the manuscript, including the abstract, introduction, methods, results, discussion, and conclusion. H.A. also ensured the accuracy and integrity of the data, managed the ethical approval process, and finalized the manuscript for submission. H.A. approved the final version of the manuscript and agrees to be accountable for all aspects of the study.

Data availability The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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